CIVICPULSE

Multimodal Civic Infrastructure Triage & Hyperlocal Home Utility Marketplace

Comprehensive Technical Architecture, Code Dossier & Judge Defense Guide

Submitted For:	Google Gemma Hackathon 2026 (Open Source & Best Use of Gemma)
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Primary AI Engine:	Gemma 4 Multimodal (gemma-4-26b-a4b-it) via google-genai SDK
Deployment Status:	Production-Ready Verified Prototype (FastAPI + SQLite WAL + Tailwind UI)
Live Evaluation Fast-Pass:	One-Click Role Switching at http://localhost:8000/#/auth

1. Executive Summary & Problem-Solution Matrix

Modern urban residents face a dual dilemma daily: **Outside the Home**, physical infrastructure hazards (open manholes, live snapped power lines, burst water pipelines, and monsoon potholes) frequently languish because citizens cannot identify whether jurisdiction lies with the Municipal Corporation, State PWD, NHAI, or Indian Railways. **Inside the Home**, residents face unregulated utility repair markets where contractors levy arbitrary visit fees and unpredictable quotes without objective benchmarks.

CivicPulse solves both problems simultaneously on a single unified platform. Leveraging Google's open-weight **Gemma 4 Multimodal AI**, CivicPulse acts as an impartial visual forensics auditor for municipal infrastructure, an automated legal jurisdiction router, and a transparent fair-price bracket estimator for domestic utility repairs.

Problem Dimension	Traditional Government Portals / Existing Apps	CivicPulse Multimodal Solution
Public Hazard Triage	Manual text forms with rigid drop-down menus; citizen must guess department; wrong routing leads to rejection.	**Gemma 4 Forensic Multimodal Vision**: Citizen snaps a photo; AI detects structural failure, identifies department, and assigns severity instantly.
Statutory Jurisdiction	Complaints on National Highways or Railways get bounced between municipal bodies with zero accountability.	**Autonomous Jurisdiction Routing**: Automatically detects non-local assets and attaches 1-click links to CPGRAMS, Sukhad Yatra, or RailMadad.

Household Repairs	Opaque contractor pricing; high aggregator commissions; zero transparency on parts vs. labor costs.	**AI Diagnosis & 2-Way Negotiation**: Gemma predicts fair cost brackets; 2-way real-time bargaining engine locks rates in escrow.
Legal Escalation	Citizens must consult lawyers or RTI activists to draft formal complaints when municipal SLAs expire.	**1-Click RTI Legal Letter Generator**: Generates formal Section 6(1) Right to Information compliance letters automatically.

2. Technology Stack & Architectural Implementation

Component Layer	Technology Used	Architectural Justification & Role
Multimodal AI	**Google Gemma 4** (gemma-4-26b-a4b-it)	Open-weight multimodal foundation model accessed via the new official google-genai SDK. Inspects raw image bytes to assess asphalt degradation, spalling concrete, pipe micro-fractures, and electrical fire hazards.
Backend Core	**FastAPI** + Uvicorn (Python 3.9+ / Pydantic v2)	High-throughput asynchronous ASGI microframework with strict schema validation, type-safe request serialization, and role-based access control (RBAC) middleware.
Data Storage	**SQLite3 WAL Mode** + Dynamic Schema Migrations	Lightweight, zero-Docker relational persistence with Write-Ahead Logging (WAL) for high concurrency, thread safety locks, and live column migrations.
Frontend Framework	**Modern HTML5 + Tailwind CSS** + Vanilla JS & Lucide Icons	Zero-bundler, ultra-responsive consumer-grade glassmorphic interface with GPU-accelerated micro-animations, laser upload scanner, and modal managers.
Audio AI (CivicTalk)	**Web Audio API** + Multimodal Audio Routing	In-browser microphone capture with live waveform visualizer, sending vernacular spoken complaints directly into multimodal audio processing.
Access Control (RBAC)	**Strict Multi-Role Gateway** (Citizen / Municipal / Technician)	Enforces strict client-side route guards (#/citizen, #/municipal, #/technician) and server-side X-User-Role header verification.

3. Deep Codebase Architecture & File Structure

File Path	Lines of Code	Key Responsibilities & Engineering Mechanisms
`backend/main.py`	~700 LOC	Entry point for ASGI server; implements require_role(...) FastAPI dependency injection; exposes REST endpoints for civic feeds, multimodal inspection, 2-way bargaining, and contractor review processing.

`backend/database.py`	~1,700 LOC	Thread-safe connection manager with db_lock; implements dynamic column PRAGMA migrations without migration scripts; handles transactional atomic operations for accepted bids and rival auto-rejections; seeds 10 multi-persona community hazards with authentic photographic assets.
`backend/models.py`	~140 LOC	Pydantic v2 schemas validating request/response models: CivicPostCreate, BidCreate, CounterBidCreate, AcceptBidCreate, and ReviewCreate.
`backend/gemma_service.py`	~350 LOC	Manages the multi-tier LLM inference pipeline: calls Gemma 4 primary target with image bytes; automatically fails over to Gemini Flash on quota spikes, and falls back to deterministic heuristic triage if network connectivity drops.
`frontend/index.html`	~1,600 LOC	Dual-view structure containing standalone #landingView and full-fledged #appWorkspaceView; hosts modal dialogs for laser upload, RTI letter generation, and star ratings.
`frontend/static/js/app.js`	~3,700 LOC	Central client-side state machine: handles hash-based RBAC routing, live bid counter-offer interaction, dual-tab social feed filtering, image zoom lightbox, and CivicTalk voice recording.
`frontend/static/css/style.css`	~850 LOC	Custom styles for glowing glassmorphism, animated holographic laser scanning line, deal confirmation glow effects, and custom scrollbars.

4. Core Functional Value Drivers & Innovations

A. Two-Way Bid & Counter-Offer Negotiation Engine with Escrow Rate Lock

Unlike Craigslist or Urban Company which force rigid, one-sided pricing, CivicPulse empowers real-time bilateral price discovery. A technician reviews a lead and submits a bid (e.g., ■550). The homeowner counter-offers (e.g., ■480). The technician can either accept or propose a revised re-counter (e.g., ■500). Once either party clicks **Accept Deal**, the database executes an atomic transaction: the agreed price is locked, *competing bids are auto-marked as rejected*, and the job transitions to 'In Progress'.

B. Dual-Tab Community Social Feed & Personal Grievance Stepper

The Citizen portal provides two distinct tabs: Community Feed (All Reports) for public civic awareness with community upvoting, and My Reported Issues which isolates the user's specific submissions. Each personal grievance features a 4-stage visual stepper (Reported $\rightarrow$ Acknowledged $\rightarrow$ In Progress $\rightarrow$ Resolved), SLA countdown timer, and field inspector notes.

C. Authentic Photographic Evidence Across 10 Diverse Civic Archetypes

Every single demo post is paired with 100% authentic, problem-specific imagery: NH-24 monsoon pothole, overflowing market compactor, dangling high-tension cable, burst water main, inactive streetlights at night, open children park manhole, collapsed sidewalk pavers, burst sewer sludge, storm-fallen tree branch, and dumped construction rubble.

D. Automated Section 6(1) RTI Legal Complaint Generator

If municipal SLAs expire without resolution, citizens can click a single button to generate a legally structured Right to Information (RTI) application citing Section 6(1) of the RTI Act 2005, complete with jurisdictional officer designations, ticket coordinates, and statutory time limits.

5. Future Architectural Scaling Roadmap

While the current system runs efficiently on a zero-docker stack (FastAPI + SQLite WAL), the following architectural evolution is engineered for national metropolitan scale deployment (10M+ citizens):

Evolution Pillar	Target Architecture	Engineering Impact & Scalability Metric
Enterprise Spatial Database	**PostgreSQL 16 + PostGIS** + TimescaleDB Telemetry	Enables sub-millisecond geospatial polygon ward matching (ST_Contains) and automated radius-based technician lead alerts (ST_DWithin). TimescaleDB stores microsecond SLA telemetry for municipal response times.
Real-Time Event Fabric	**Redis 7 Pub/Sub Cluster** + FastAPI WebSockets	Replaces client-side HTTP polling with an event-driven pub/sub bus. Provides sub-50ms live bidding updates, instant counter-offer notifications, and live GPS tracking of dispatched municipal repair trucks.
Payment & Escrow Engine	**Razorpay Route / UPI Intent** + RBI Smart Escrow	When a resident accepts a quote, funds are locked in an automated escrow mandate. Funds are released to the technician only after dual-party OTP confirmation or GPS geofenced work verification.
Edge Multimodal AI	**On-Device Gemma 4 INT4** via Google MediaPipe	Runs quantized 4-bit Gemma models locally on low-cost Android smartphones. Enables rural citizens in zero-connectivity areas to audit broken borewells and roads offline, auto-syncing when network reconnects.
National GovTech Hooks	**CPGRAMS & Swachhata APIs** + DigiLocker / Aadhaar e-KYC	Bi-directional integration with Central Government grievance databases and Aadhaar e-KYC to eliminate fraudulent complaint spamming while verifying technician professional licenses.

6. How to Explain CivicPulse to Judges (Winning Defense)

The 90-Second Elevator Pitch (Formal Verbatim Script):

"Distinguished judges, every urban resident in developing nations faces two deeply frustrating problems every day: Outside our homes, broken municipal infrastructure sits neglected because citizens get trapped in administrative red-tape between local municipal corporations, national highway authorities, and railways. Inside our homes, private utility repairs are unregulated, leaving families vulnerable to arbitrary contractor quotes and price-gouging.

We built CivicPulse to solve both problems with a single, open-weight multimodal intelligence engine. Powered by Google's Gemma 4, a citizen simply snaps a photograph. Gemma 4 performs forensic structural visual analysis, assigns objective hazard severity, and autonomously routes the ticket to the exact jurisdictional body—whether local municipality or apex central portals like CPGRAMS. For domestic repairs, Gemma diagnoses the defect and computes fair market labor brackets, feeding into our real-time two-way bargaining engine where homeowners and verified technicians negotiate and lock rates in escrow. With strict role-based access control, multi-stage SLA tracking, and one-click statutory RTI letter generation, CivicPulse transforms passive complainers into active civic stakeholders."

Anticipated Tough Judge Questions & Bulletproof Formal Answers:

Judge Q1: Why did you use Gemma 4 instead of a traditional drop-down text form?

Answer: Traditional drop-down forms are the number-one reason civic grievance systems fail. When an untrained citizen sees a cracked road, they don't know if it's sub-base soil erosion, asphalt shearing, or a water supply pipeline leak underneath. Gemma 4 processes raw visual evidence multimodally, detecting structural fault physics that no drop-down form can capture, while completely eliminating user categorization errors.

Judge Q2: How does your bargaining engine prevent price gouging or lowball exploitation?

Answer: CivicPulse introduces Gemma 4 as an impartial algorithmic anchor. Before any bids are placed, Gemma computes a transparent 'Fair Market Benchmark' based on prevailing labor and materials indices across Indian cities. Homeowners and technicians negotiate with complete visibility of this benchmark, preventing predatory price surges while ensuring technicians earn a dignified, fair wage.

Judge Q3: Municipal corporations are notoriously slow. How does CivicPulse enforce accountability?

Answer: We implement a 4-stage transparent lifecycle stepper with live SLA countdown timers and public comment threads. Crucially, if an authority breaches statutory SLA deadlines, CivicPulse automatically generates a legally valid Section 6(1) Right to Information (RTI) application citing officer designations and GPS coordinates, creating immediate statutory legal accountability.

Judge Q4: How is CivicPulse architected for real-world production scale?

Answer: Our architecture enforces clean separation of concerns: a stateless FastAPI ASGI core, dynamic SQLite WAL transactional persistence with zero downtime, and a multi-tiered AI fallback (Gemma 4 primary $\rightarrow$ Gemini Flash cloud $\rightarrow$ offline heuristics). For enterprise scale, our modular schema is 100% prepared to migrate to PostgreSQL with PostGIS geofencing and Redis Pub/Sub WebSockets.

Summary for Evaluators: *CivicPulse was designed, engineered, and verified under real-world conditions. Judges are invited to visit http://localhost:8000/#/auth to test the live system via our ■ Evaluator Fast-Pass.*